

# Direct Preference Optimization

DATA 443/643 · Week 9, note 9.3

Week 9

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Ideas from Stanford CS224N 2025 L10 (DPO: write  $r$  from  $\pi$ ;  $Z(x)$  cancels). Original slides; that course is not copied.

# Lecture plan

- 1 Skip the RM and the PPO loop
- 2 Closed-form  $r$  from the policy
- 3 Init loss is still  $\log 2$
- 4 Offline: no new  $y$  unless you collect them

Key goal: the frozen reference is the leash. Dropping it is not DPO.

# First time: what / why / formula

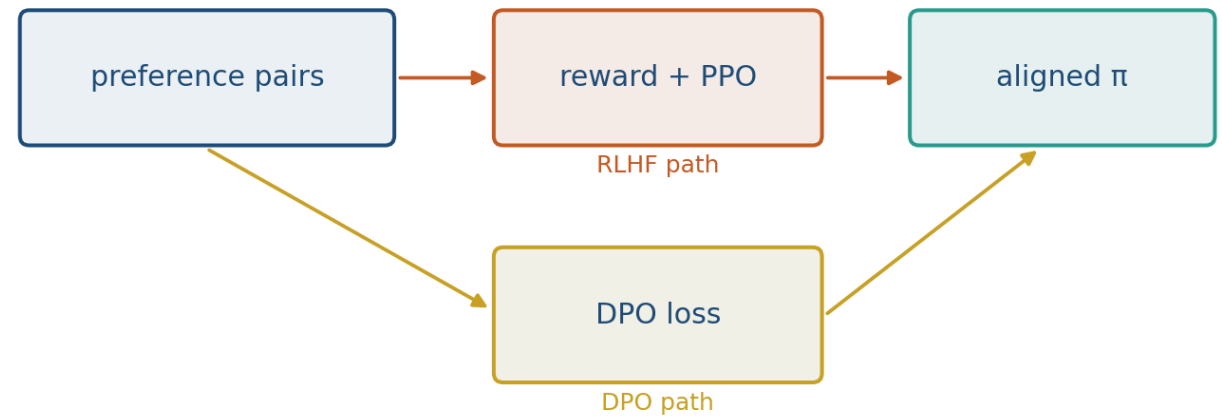
- What / why. Under Bradley–Terry, the optimal reward is  $r = \beta \log(\pi / \pi_{\text{ref}}) + c(x)$ . Plug into the logistic loss;  $r$  disappears. Train  $\pi_\theta$  offline. RLHF's PPO stage is heavy. DPO is a classification loss on logits of  $y_w$  and  $y_l$ .
- Architecture. Policy + frozen reference. No critic, no sampling-in-the-loop required (you need logprobs of both completions).
- Formula.  $\mathcal{L}_{\text{DPO}} = -\log \sigma \left( \beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right)$ .
- Tradeoffs. + Offline, no RM. – Still needs pairs and a reference; can overfit; not magic if the pairs are noisy.

How. Lab 9: toy scalar logits. Start:  $\pi = \pi_{\text{ref}} \Rightarrow \text{loss } \log 2$ .

# Same pairs, shorter stack

- CS224N L10: the optimal  $r$  is a log-ratio plus  $Z(x)$ .
- $Z(x)$  cancels in a pairwise difference.
- KTO and IPO exist. This hour is DPO.

DPO fits the same pairwise data without an explicit reward model



# The implicit reward

$$r(x, y) = \beta \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

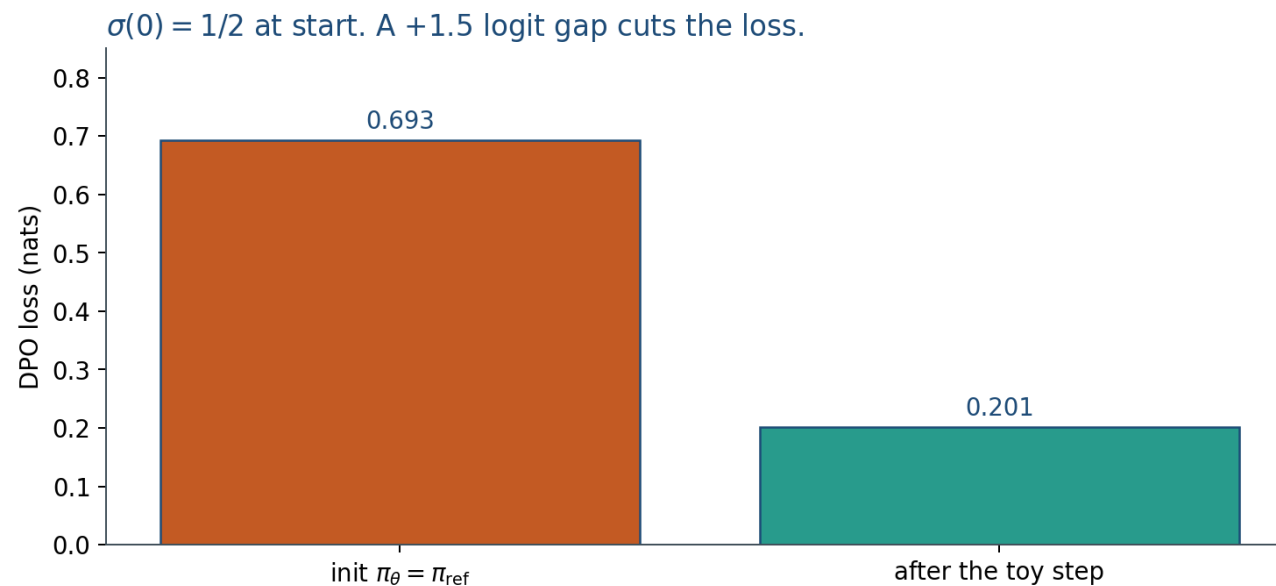
Plug into Bradley–Terry. The  $\mathcal{R}$  head disappears.

If  $\pi_{\theta} = \pi_{\text{ref}}$ , the argument of  $\sigma$  is 0.

Trainable object is  $\pi_{\theta}$ . The reference stays frozen.

# A toy step

- Init:  $\mathcal{L} = \log 2 \approx 0.693$ .
- Winner log-ratio 0.5, loser  $-1.0$ , gap 1.5.
- $\sigma(1.5) \approx 0.818$ , loss  $\approx 0.201$ .



# What the pair file cannot do

## PPO

Samples new  $y$  from the current policy. Can explore replies that were never labeled.

## Offline DPO

Fixed pairs. If SFT never wrote a good  $y_w$ , DPO cannot invent it.

# What to take from this lecture

## Init

$\sigma(0)=1/2$  is the start, not a bug.

## Report

$\beta$ , the reference checkpoint, held-out preference accuracy.

## Lab 9

Toy logits. Not an LLM PPO run.